

Enterprise Traffic Flow: Beginner-Friendly Explanation

1. The user enters the application URL

A user opens a browser and enters something like **API . company . com**. This is simply the name people use to reach our application.

2. Route 53 finds where the application lives

Route 53 is like the Internet's phonebook. It takes **API . company . com** and finds the address of our Application Load Balancer.

The user does not need to know where the actual application servers are.

3. Cloudflare can provide an extra security layer

In some enterprises, the request first goes through **Cloudflare**. It can check for things such as suspicious traffic, attacks, bots, and DDoS activity before allowing the request to continue.

4. The request enters AWS through the Internet Gateway

The request now reaches our AWS network.

The **Internet Gateway** is the connection between our VPC and the public Internet.

5. The Application Load Balancer receives the request

The **ALB** is the main entry point for the application.

It receives the request and asks, essentially, "**Which healthy application should receive this request?**"

6. The ALB sends the request to an EKS pod

The ALB forwards the request to a healthy **EKS pod**.

A pod is where our application container is actually running.

7. The application runs inside private subnets

Our application servers and pods are kept in **private subnets**.

Users on the Internet **cannot connect directly to these pods**. They must come through the controlled entry point, such as the ALB.

8. The pod processes the request and sends the response back

The application processes the user's request and sends the response back through the networking path, ultimately reaching the user.

9. What happens when the application needs the Internet?

Suppose our application needs to call an external API, send a notification, or retrieve information from another service.

The **pod starts the outbound connection**.

10. The request goes to a NAT Gateway

Because the pod is private, it cannot directly communicate with the public Internet. Its route table sends Internet-bound traffic to a **NAT Gateway**.

11. NAT Gateway provides the public identity

The NAT Gateway replaces the pod's private IP address with a public **Elastic IP address**. The external service therefore sees the **NAT Gateway's public IP**, not the pod's private IP.

12. The outbound request leaves AWS through the Internet Gateway

The NAT Gateway sends the request toward the Internet through the **Internet Gateway**, and the external service receives it.

13. Why do we use two Availability Zones?

We don't want the entire application to depend on one physical location.

So we spread resources across **multiple Availability Zones**. If one AZ has a failure, the ALB can continue sending traffic to healthy resources in another AZ.

14. Why are private and public subnets used?

Think of it as security boundaries.

Public subnets contain components that need Internet-facing connectivity, while **private subnets** contain application workloads that should not be directly reachable from the Internet.

15. What about a complete AWS Region failure?

For larger enterprises, we can have a **Disaster Recovery environment in another AWS Region**.

Route 53 can then direct users toward the recovery environment when the primary region is unavailable.

16. Two important disaster recovery terms are RTO and RPO

RTO means: *How quickly do we need the application back?*

RPO means: *How much recent data can we afford to lose?*

The easiest way to remember the whole flow

Incoming traffic:

User → Route 53 → Cloudflare (optional) → Internet Gateway → ALB → EKS Pod

Outgoing traffic from the application:

EKS Pod → Route Table → NAT Gateway → Internet Gateway → External Service

It can apply to **both public and private applications**, but the **entry point and network path are different**.

1. Public application

For an Internet-facing application, the typical flow is:

User on Internet → Route 53 → Cloudflare (optional) → Internet Gateway → Internet-facing ALB → EKS Pod

The important point is that the **ALB is reachable from the Internet**, while the EKS pods can still remain in private subnets.

Example:

api.company.com is a public API used by customers.

2. Private application

For an internal application, users generally **do not come from the public Internet**.

A common enterprise flow is:

Employee / Internal Service → VPN or Direct Connect / Transit Gateway → Internal ALB → EKS Pod

The ALB would typically be an **internal ALB**, not an Internet-facing ALB.

Example:

internal-api.company.local is used only by applications or employees inside the corporate network.

3. The EKS side can look very similar

This is the important concept:

Whether the application is public or private, the **application pods can still be in private subnets**.

The difference is mainly **how traffic reaches the ALB**:

Public Application	Private Application
Internet users	Employees/internal systems
Internet-facing ALB	Internal ALB
Internet Gateway involved for inbound traffic	Usually private connectivity such as VPN, Direct Connect, Transit Gateway
Cloudflare may be used	Usually not required for Internet protection
Pods can remain private	Pods can remain private

4. Outbound traffic is a separate story

The **NAT Gateway concept is primarily about outbound Internet access** from private resources.

For example:

Private EKS Pod → NAT Gateway → Internet Gateway → External API

This can happen whether the application itself is **public or private**.

So don't associate **NAT Gateway = public application**.

They solve different problems:

- **ALB** = how clients reach the application
- **NAT Gateway** = how private resources reach the Internet

The easiest way to explain it

Public or private describes who can access the application. Private subnet describes where the application resources live.

So an enterprise can have:

Public application:

Internet → Internet-facing ALB → Private EKS Pods

Private application:

Corporate Network → Internal ALB → Private EKS Pods

And in both cases:

EKS Pods → NAT Gateway → Internet, when the pods need outbound Internet access.